

Name _____ Period _____

Skill 6.01 Exercise 1

Consider a computer that uses 6 bits to represent integers: 1 bit for the sign and 5 bits for the actual number. What's the largest positive integer it can represent?

If the left-most bit represents the sign, where 0 = positive and 1 = negative, what is 11001 in decimal?

11001

If the left-most bit represents the sign, where 0 = positive and 1 = negative, what is -21 in decimal?

Skill 6.02 Exercise 1

On a computer which uses 6 bits to represent integers (with 1 bit to represent the sign), which of these operations result in overflow?

15+15

8 * 4

3+29

30+1

Skill 6.03 Exercise 1

Convert each of the floating point numbers to binary

0.125

0.03125

0.675

0.375

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10.75

Skill 6.04 Exercise 1

A scientist is running a program to calculate the volume of a cone:

```
radius ← 17.24  
height ← 5.24  
volume ← PI * (radius * radius) * (height / 3)
```

The code relies on the built-in constant PI. After running the code, the variable volume stores 1630.9266447568566.

Their supervisor checks their results by running the same calculation on their own computer. Their program results in a volume of 1630.9266447564448.

The two values are very close, but not quite the same. Why?

Skill 6.04 Exercise 2

Nuru writes this code to calculate the final cost of an item with a discount applied:

```
price ← 0.7  
discount ← 0.2  
final ← price - discount
```

They're surprised to see that final stores the value 0.49999999999999994 instead of 0.5.

What is the best explanation for that result?